

Harsh Jain

Tucson, AZ | harshjain.cs.30@gmail.com | +1(520)551-1304 | linkedin.com/in/-harshitjain | github.com/Harsh7115 | Portfolio

Experience

AI Systems Engineer (POC), University of Arizona × McGraw Hill, Remote **Jan 2026 – Present**

- Built evaluation pipelines to identify hallucination patterns and reasoning failures across multi-step LLM outputs.
- Developed and evaluated prompt chaining workflows with intermediate quality checks to improve LLM output reliability.
- Designed systematic experiments to measure capability gaps in domain-specific reasoning, documenting failure modes and mitigations.
- Architected a content-grounded RAG system to constrain model behavior under domain-specific contexts.
- Investigated how retrieval design and chunking strategies influence factual consistency and failure modes under distribution shifts.

Computer Science Fellow, Handshake AI MOVE Program, Remote **Sep 2025 – Present**

- Evaluated 100+ LLM outputs weekly to identify reasoning failures and alignment issues across diverse prompt distributions.
- Analyzed agent failure patterns and contributed to improving evaluation benchmarks through iterative feedback loops.
- Authored adversarial and fail-to-pass test cases to surface edge-case failures in AI software engineering agents.
- Built reproducible Docker environments to enable consistent large-scale evaluation across model variants.

GRC Business Analyst, University of Arizona ITS, Tucson **Sep 2023 – Aug 2025**

- Evaluated AI-assisted tools like ServiceNow Now Assist in production, identifying failure modes, edge cases, and system limitations.
- Analyzed system behavior under varied, adversarial, and distribution-shifted inputs to characterize robustness and failure modes.
- Developed structured testing workflows to validate AI outputs across compliance-sensitive environments.

Founding Engineer, SkillfullyAware (SAAQ) App, Tucson **Oct 2025 – Jan 2026**

- Prototyped LLM-powered features end-to-end, systematically evaluating output quality and identifying failure patterns under real usage conditions.
- Integrated LLM APIs and vector retrieval pipelines, iterating on design based on empirical quality assessment across model variants.
- Defined evaluation criteria for LLM integration decisions, documenting behavioral trade-offs and reliability gaps across approaches.

AI Solutions Developer, Andrew Weil Center for Integrative Medicine, Tucson **Aug 2025 – Nov 2025**

- Maintained reliable AWS Lambda-based pipelines within a regulated healthcare environment, emphasizing data integrity.
- Evaluated AI component behavior against clinical requirements, identifying edge cases that affected output reliability.
- Collaborated with clinical and technical teams to ensure production-grade safety and reliability of deployed systems.

Projects

Guardrail Auditor guardrail-auditor.vercel.app

- **Tools:** Next.js 14, TypeScript, Prisma, Neon Postgres, Vercel
- Developed a framework to systematically characterize LLM robustness against prompt injection, role bypass, data leakage, and grounding failures.
- Designed risk-tiered scoring to quantify safety violations; explored how adversarial inputs shift model behavior across attack vectors.
- Enabled structured failure mode reporting with exportable metrics (JSON/CSV) to support reproducible analysis and iterative evaluation.

go-distributed-kv github.com/Harsh7115/go-distributed-kv

- **Tools:** Go, Raft Consensus, gRPC, Docker
- Implemented a distributed key-value store with Raft consensus for leader election and log replication, achieving linearizable reads and fault-tolerant writes validated under simulated node failures and network partitions.

c-coroutine github.com/Harsh7115/c-coroutine

- **Tools:** C, x86-64 Assembly, POSIX
- Cooperative userspace fiber library in C with hand-written x86-64 context switching (System V AMD64 ABI), a FIFO scheduler, and `co_yield/co_await` primitives; compile-time selection between `asm` and `ucontext` backends.

Technologies

Languages: Python, Go, Java, C, C++, TypeScript, SQL, JavaScript

AI Safety & Evaluation: LLM Evaluation, Adversarial Testing, AI Guardrails, Prompt Injection, RAG Safety, Claude, Claude Code

Systems & Infrastructure: RAG Pipelines, Docker, AWS (Lambda, S3), Distributed Systems, gRPC, PostgreSQL, Linux

Engineering Practices: System Design, Empirical Evaluation, Agile, TDD, Code Review

Education

University of Arizona, BS in Computer Science, Minor in Software Engineering **Aug 2022 – May 2026**

- GPA and Honors: 3.85/4.0 (Highest Merit Scholarship, Dean's List, Highest Academic Distinction)
- Coursework: Data Structures & Algorithms, Object Oriented Programming, Data Science & Neural Networks, Advanced C Programming, Computer Architecture, Software Engineering, Operating Systems, Computer Networking
- Research focus: LLM evaluation pipeline design, adversarial test case authoring, AI safety and robustness engineering